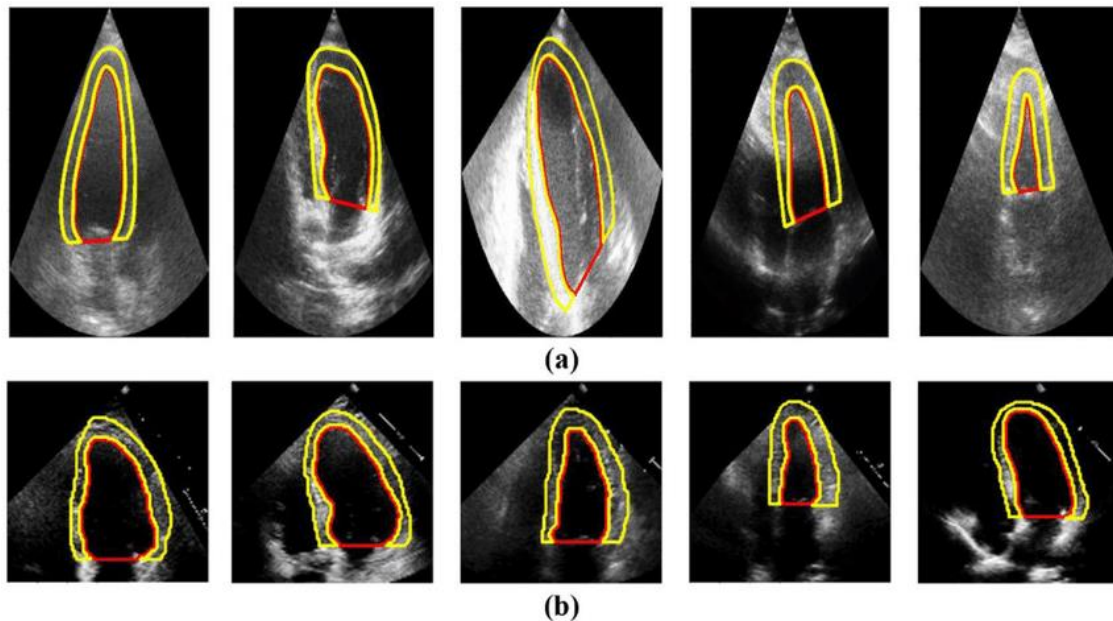


CAMUS-Human Heart Data Segmentation



```
import numpy as np
import pandas as pd
import os

import cv2
import matplotlib.pyplot as plt

image_dir = '/kaggle/input/camus-echocardiography-image-dataset/frames/'
mask_dir = '/kaggle/input/camus-echocardiography-image-dataset/masks/'

if not os.path.exists(image_dir):
    raise FileNotFoundError(f"Image directory {image_dir} does not exist")
if not os.path.exists(mask_dir):
    raise FileNotFoundError(f"Mask directory {mask_dir} does not exist")

image_files = [f for f in os.listdir(image_dir) if f.endswith('.png')]
mask_files = [f for f in os.listdir(mask_dir) if f.endswith('.png')]

print(f"Found {len(image_files)} image files in {image_dir}")
print(f"Found {len(mask_files)} mask files in {mask_dir}")

image_basenames = {os.path.splitext(f)[0].replace('frame_', ''): f for f in image_files}
mask_basenames = {os.path.splitext(f)[0].replace('mask_', ''): f for f in mask_files}
```

```

print("Sample image basenames:", list(image_basenames.keys())[:5])
print("Sample mask basenames:", list(mask_basenames.keys())[:5])

common_keys = set(image_basenames.keys()) & set(mask_basenames.keys())

print(f"Found {len(common_keys)} matching image-mask pairs")

data = []
for key in sorted(common_keys):
    image_path = os.path.join(image_dir, image_basenames[key])
    mask_path = os.path.join(mask_dir, mask_basenames[key])
    data.append({'image_path': image_path, 'mask_path': mask_path})

df = pd.DataFrame(data)

```

```

Found 900 image files in /kaggle/input/camus-echocardiography-image-
dataset/frames/
Found 900 mask files in /kaggle/input/camus-echocardiography-image-
dataset/masks/
Sample image basenames: ['0221', '0289', '0790', '0140', '0537']
Sample mask basenames: ['0595', '0170', '0166', '0570', '0134']
Found 900 matching image-mask pairs

```

```
df.head()
```

```

                                     image_path \
0  /kaggle/input/camus-echocardiography-image-dat...
1  /kaggle/input/camus-echocardiography-image-dat...
2  /kaggle/input/camus-echocardiography-image-dat...
3  /kaggle/input/camus-echocardiography-image-dat...
4  /kaggle/input/camus-echocardiography-image-dat...

                                     mask_path
0  /kaggle/input/camus-echocardiography-image-dat...
1  /kaggle/input/camus-echocardiography-image-dat...
2  /kaggle/input/camus-echocardiography-image-dat...
3  /kaggle/input/camus-echocardiography-image-dat...
4  /kaggle/input/camus-echocardiography-image-dat...

```

```
df.tail()
```

```

                                     image_path \
895 /kaggle/input/camus-echocardiography-image-dat...
896 /kaggle/input/camus-echocardiography-image-dat...
897 /kaggle/input/camus-echocardiography-image-dat...
898 /kaggle/input/camus-echocardiography-image-dat...
899 /kaggle/input/camus-echocardiography-image-dat...

                                     mask_path

```

```

895 /kaggle/input/camus-echocardiography-image-dat...
896 /kaggle/input/camus-echocardiography-image-dat...
897 /kaggle/input/camus-echocardiography-image-dat...
898 /kaggle/input/camus-echocardiography-image-dat...
899 /kaggle/input/camus-echocardiography-image-dat...

df.shape

(900, 2)

df.columns

Index(['image_path', 'mask_path'], dtype='object')

df.duplicated().sum()

0

df.isnull().sum()

image_path    0
mask_path     0
dtype: int64

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 900 entries, 0 to 899
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   image_path  900 non-null    object
 1   mask_path   900 non-null    object
dtypes: object(2)
memory usage: 14.2+ KB

import warnings
warnings.filterwarnings('ignore')

def display_image_mask_pairs(df, num_display=5):

    num_display = min(num_display, len(df))
    if num_display == 0:
        print("No image-mask pairs to display.")
        return

    display_df = df.head(num_display)

    fig, axes = plt.subplots(2, num_display, figsize=(5 * num_display, 6))

    if num_display == 1:
        axes = [[axes[0]], [axes[1]]]

```

```

for i in range(num_display):
    image = cv2.imread(display_df['image_path'].iloc[i],
cv2.IMREAD_GRAYSCALE)
    mask = cv2.imread(display_df['mask_path'].iloc[i],
cv2.IMREAD_GRAYSCALE)

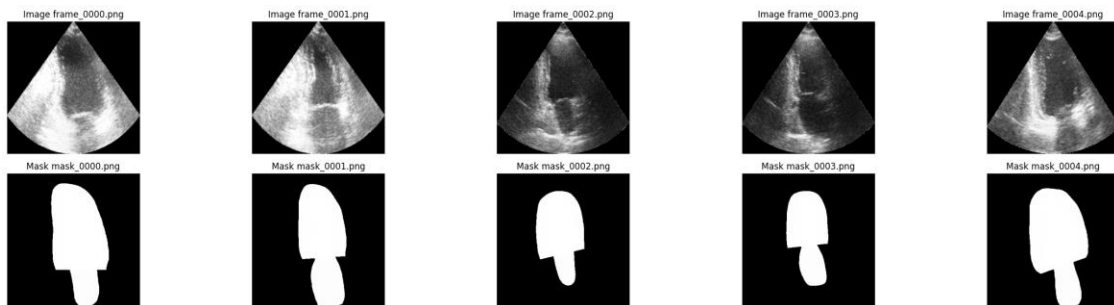
    axes[0][i].imshow(image, cmap='gray')
    axes[0][i].set_title(f'Image
{os.path.basename(display_df["image_path"].iloc[i])}')
    axes[0][i].axis('off')

    axes[1][i].imshow(mask, cmap='gray')
    axes[1][i].set_title(f'Mask
{os.path.basename(display_df["mask_path"].iloc[i])}')
    axes[1][i].axis('off')

plt.tight_layout()
plt.show()

display_image_mask_pairs(df, num_display=5)

```



```

from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow.keras import layers, models

def dice_coefficient(y_true, y_pred, smooth=1e-6):
    y_true_f = tf.keras.backend.flatten(y_true)
    y_pred_f = tf.keras.backend.flatten(y_pred)
    intersection = tf.keras.backend.sum(y_true_f * y_pred_f)
    return (2. * intersection + smooth) / (tf.keras.backend.sum(y_true_f) +
tf.keras.backend.sum(y_pred_f) + smooth)

def iou(y_true, y_pred, smooth=1e-6):
    y_true_f = tf.keras.backend.flatten(y_true)
    y_pred_f = tf.keras.backend.flatten(y_pred)
    intersection = tf.keras.backend.sum(y_true_f * y_pred_f)
    union = tf.keras.backend.sum(y_true_f) + tf.keras.backend.sum(y_pred_f) -
intersection
    return (intersection + smooth) / (union + smooth)

```

```

def unet_model(input_shape=(256, 256, 1)):
    inputs = layers.Input(input_shape)

    c1 = layers.Conv2D(64, 3, activation='relu', padding='same')(inputs)
    c1 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1)
    p1 = layers.MaxPooling2D((2, 2))(c1)

    c2 = layers.Conv2D(128, 3, activation='relu', padding='same')(p1)
    c2 = layers.Conv2D(128, 3, activation='relu', padding='same')(c2)
    p2 = layers.MaxPooling2D((2, 2))(c2)

    c3 = layers.Conv2D(256, 3, activation='relu', padding='same')(p2)
    c3 = layers.Conv2D(256, 3, activation='relu', padding='same')(c3)
    p3 = layers.MaxPooling2D((2, 2))(c3)

    c4 = layers.Conv2D(512, 3, activation='relu', padding='same')(p3)
    c4 = layers.Conv2D(512, 3, activation='relu', padding='same')(c4)
    p4 = layers.MaxPooling2D((2, 2))(c4)

    c5 = layers.Conv2D(1024, 3, activation='relu', padding='same')(p4)
    c5 = layers.Conv2D(1024, 3, activation='relu', padding='same')(c5)

    u6 = layers.Conv2DTranspose(512, 2, strides=(2, 2), padding='same')(c5)
    u6 = layers.concatenate([u6, c4])
    c6 = layers.Conv2D(512, 3, activation='relu', padding='same')(u6)
    c6 = layers.Conv2D(512, 3, activation='relu', padding='same')(c6)

    u7 = layers.Conv2DTranspose(256, 2, strides=(2, 2), padding='same')(c6)
    u7 = layers.concatenate([u7, c3])
    c7 = layers.Conv2D(256, 3, activation='relu', padding='same')(u7)
    c7 = layers.Conv2D(256, 3, activation='relu', padding='same')(c7)

    u8 = layers.Conv2DTranspose(128, 2, strides=(2, 2), padding='same')(c7)
    u8 = layers.concatenate([u8, c2])
    c8 = layers.Conv2D(128, 3, activation='relu', padding='same')(u8)
    c8 = layers.Conv2D(128, 3, activation='relu', padding='same')(c8)

    u9 = layers.Conv2DTranspose(64, 2, strides=(2, 2), padding='same')(c8)
    u9 = layers.concatenate([u9, c1])
    c9 = layers.Conv2D(64, 3, activation='relu', padding='same')(u9)
    c9 = layers.Conv2D(64, 3, activation='relu', padding='same')(c9)

    outputs = layers.Conv2D(1, 1, activation='sigmoid')(c9)

    model = models.Model(inputs, outputs)
    return model

def load_and_preprocess_data(image_dir, mask_dir, img_size=(256, 256)):

```

```

image_files = [f for f in os.listdir(image_dir) if f.endswith('.png')]
mask_files = [f for f in os.listdir(mask_dir) if f.endswith('.png')]

print(f"Found {len(image_files)} image files in {image_dir}")
print(f"Found {len(mask_files)} mask files in {mask_dir}")

image_basenames = {os.path.splitext(f)[0].replace('frame_', ''): f for f
in image_files}
mask_basenames = {os.path.splitext(f)[0].replace('mask_', ''): f for f in
mask_files}

common_keys = set(image_basenames.keys()) & set(mask_basenames.keys())
print(f"Found {len(common_keys)} matching image-mask pairs")

images = []
masks = []
for key in sorted(common_keys):
    img_path = os.path.join(image_dir, image_basenames[key])
    mask_path = os.path.join(mask_dir, mask_basenames[key])

    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    mask = cv2.imread(mask_path, cv2.IMREAD_GRAYSCALE)

    img = cv2.resize(img, img_size)
    mask = cv2.resize(mask, img_size)

    img = img.astype(np.float32) / 255.0
    mask = (mask > 0).astype(np.float32)

    images.append(img)
    masks.append(mask)

images = np.array(images)[..., np.newaxis]
masks = np.array(masks)[..., np.newaxis]

return images, masks

def plot_training_history(history):
    fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(15, 5))

    ax1.plot(history.history['loss'], label='Train Loss')
    ax1.plot(history.history['val_loss'], label='Val Loss')
    ax1.set_title('Loss')
    ax1.set_xlabel('Epoch')
    ax1.set_ylabel('Loss')
    ax1.legend()

    ax2.plot(history.history['dice_coefficient'], label='Train Dice')

```

```

ax2.plot(history.history['val_dice_coefficient'], label='Val Dice')
ax2.set_title('Dice Coefficient')
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Dice')
ax2.legend()

ax3.plot(history.history['iou'], label='Train IoU')
ax3.plot(history.history['val_iou'], label='Val IoU')
ax3.set_title('IoU')
ax3.set_xlabel('Epoch')
ax3.set_ylabel('IoU')
ax3.legend()

plt.tight_layout()
plt.show()

def display_predictions(model, x_test, y_test, num_display=5):
    num_display = min(num_display, len(x_test))
    if num_display == 0:
        print("No test samples to display.")
        return

    preds = model.predict(x_test[:num_display])
    preds = (preds > 0.5).astype(np.float32) # Binarize predictions

    fig, axes = plt.subplots(3, num_display, figsize=(5 * num_display, 7))
    if num_display == 1:
        axes = [[axes[0]], [axes[1]], [axes[2]]]

    for i in range(num_display):

        axes[0][i].imshow(x_test[i, :, :, 0], cmap='gray')
        axes[0][i].set_title(f'Image {i+1}')
        axes[0][i].axis('off')

        axes[1][i].imshow(y_test[i, :, :, 0], cmap='gray')
        axes[1][i].set_title(f'Ground Truth {i+1}')
        axes[1][i].axis('off')

        axes[2][i].imshow(preds[i, :, :, 0], cmap='gray')
        axes[2][i].set_title(f'Prediction {i+1}')
        axes[2][i].axis('off')

    plt.tight_layout()
    plt.show()

image_dir = '/kaggle/input/camus-echocardiography-image-dataset/frames/'

```

```
mask_dir = '/kaggle/input/camus-echocardiography-image-dataset/masks/'
```

```
images, masks = load_and_preprocess_data(image_dir, mask_dir)
```

```
x_train, x_temp, y_train, y_temp = train_test_split(images, masks,  
test_size=0.3, random_state=42)
```

```
x_val, x_test, y_val, y_test = train_test_split(x_temp, y_temp,  
test_size=0.5, random_state=42)
```

```
print(f"Train: {len(x_train)}, Validation: {len(x_val)}, Test:  
{len(x_test)}")
```

```
model = unet_model()
```

```
model.compile(optimizer='adam', loss='binary_crossentropy',  
metrics=[dice_coefficient, iou])
```

```
history = model.fit(x_train, y_train, validation_data=(x_val, y_val),  
epochs=20, batch_size=16, verbose=1)
```

```
test_scores = model.evaluate(x_test, y_test, verbose=0)
```

```
print(f"Test Loss: {test_scores[0]:.4f}")
```

```
print(f"Test Dice Coefficient: {test_scores[1]:.4f}")
```

```
print(f"Test IoU: {test_scores[2]:.4f}")
```

```
plot_training_history(history)
```

```
display_predictions(model, x_test, y_test, num_display=5)
```

```
2025-08-05 06:54:31.486098: E
```

```
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:477] Unable to  
register cuFFT factory: Attempting to register factory for plugin cuFFT when  
one has already been registered
```

```
WARNING: All log messages before absl::InitializeLog() is called are written  
to STDERR
```

```
E0000 00:00:1754376871.865357      36 cuda_dnn.cc:8310] Unable to register  
cuDNN factory: Attempting to register factory for plugin cuDNN when one has  
already been registered
```

```
E0000 00:00:1754376871.975548      36 cuda_blas.cc:1418] Unable to register  
cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has  
already been registered
```

```
Found 900 image files in /kaggle/input/camus-echocardiography-image-  
dataset/frames/
```

```
Found 900 mask files in /kaggle/input/camus-echocardiography-image-  
dataset/masks/
```

```
Found 900 matching image-mask pairs
```

```
Train: 630, Validation: 135, Test: 135
```

```
I0000 00:00:1754376908.678064      36 gpu_device.cc:2022] Created device  
/job:localhost/replica:0/task:0/device:GPU:0 with 13942 MB memory: ->
```

```
device: 0, name: Tesla T4, pci bus id: 0000:00:04.0, compute capability: 7.5
```


I0000 00:00:1754376908.678766 36 gpu_device.cc:2022] Created device
/job:localhost/replica:0/task:0/device:GPU:1 with 13942 MB memory: ->
device: 1, name: Tesla T4, pci bus id: 0000:00:05.0, compute capability: 7.5

Epoch 1/20

WARNING: All log messages before absl::InitializeLog() is called are written
to STDERR

I0000 00:00:1754376920.247068 120 service.cc:148] XLA service
0x7fa8180092e0 initialized for platform CUDA (this does not guarantee that
XLA will be used). Devices:
I0000 00:00:1754376920.248680 120 service.cc:156] StreamExecutor device
(0): Tesla T4, Compute Capability 7.5
I0000 00:00:1754376920.248718 120 service.cc:156] StreamExecutor device
(1): Tesla T4, Compute Capability 7.5
I0000 00:00:1754376921.335422 120 cuda_dnn.cc:529] Loaded cuDNN version
90300

2025-08-05 06:55:27.406433: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng12{k11=0} for conv (f32[16,128,128,128]{3,2,1,0}, u8[0]{0}) custom-
call(f32[16,128,128,128]{3,2,1,0}, f32[128,128,3,3]{3,2,1,0}, f32[128]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]}} is
taking a while...

2025-08-05 06:55:27.518790: E
external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.112457241s
Trying algorithm eng12{k11=0} for conv (f32[16,128,128,128]{3,2,1,0},
u8[0]{0}) custom-call(f32[16,128,128,128]{3,2,1,0},
f32[128,128,3,3]{3,2,1,0}, f32[128]{0}), window={size=3x3 pad=1_1x1_1},
dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]}} is
taking a while...

I0000 00:00:1754376985.537135 120 device_compiler.h:188] Compiled cluster
using XLA! This line is logged at most once for the lifetime of the process.

39/40 ————— 0s 778ms/step - dice_coefficient: 0.3499 - iou:
0.2143 - loss: 0.5545

E0000 00:00:1754377029.049251 120 gpu_timer.cc:82] Delay kernel timed
out: measured time has sub-optimal accuracy. There may be a missing warmup
execution, please investigate in Nsight Systems.

E0000 00:00:1754377029.276821 120 gpu_timer.cc:82] Delay kernel timed
out: measured time has sub-optimal accuracy. There may be a missing warmup
execution, please investigate in Nsight Systems.

E0000 00:00:1754377035.249075 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

E0000 00:00:1754377035.560216 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

E0000 00:00:1754377043.601536 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

E0000 00:00:1754377043.838599 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

40/40 ————— 0s 2s/step - dice_coefficient: 0.3511 - iou: 0.2152 - loss: 0.5526

E0000 00:00:1754377097.407637 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

E0000 00:00:1754377097.636084 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

40/40 ————— 188s 3s/step - dice_coefficient: 0.3522 - iou: 0.2160 - loss: 0.5507 - val_dice_coefficient: 0.4524 - val_iou: 0.2924 - val_loss: 0.4079

Epoch 2/20

40/40 ————— 34s 863ms/step - dice_coefficient: 0.4051 - iou: 0.2559 - loss: 0.4737 - val_dice_coefficient: 0.4201 - val_iou: 0.2660 - val_loss: 0.4179

Epoch 3/20

40/40 ————— 34s 856ms/step - dice_coefficient: 0.4387 - iou: 0.2811 - loss: 0.4203 - val_dice_coefficient: 0.4499 - val_iou: 0.2903 - val_loss: 0.4172

Epoch 4/20

40/40 ————— 34s 857ms/step - dice_coefficient: 0.4427 - iou: 0.2847 - loss: 0.4153 - val_dice_coefficient: 0.4489 - val_iou: 0.2895 - val_loss: 0.4105

Epoch 5/20

40/40 ————— 34s 855ms/step - dice_coefficient: 0.4423 - iou: 0.2844 - loss: 0.4111 - val_dice_coefficient: 0.4513 - val_iou: 0.2915 - val_loss: 0.4143

Epoch 6/20

40/40 ————— 34s 849ms/step - dice_coefficient: 0.4471 - iou: 0.2880 - loss: 0.4068 - val_dice_coefficient: 0.4657 - val_iou: 0.3036 - val_loss: 0.3944

Epoch 7/20

40/40 ————— 34s 854ms/step - dice_coefficient: 0.4592 - iou: 0.2981 - loss: 0.3931 - val_dice_coefficient: 0.4739 - val_iou: 0.3106 - val_loss: 0.3936

Epoch 8/20

40/40 ————— 34s 862ms/step - dice_coefficient: 0.4658 - iou:
0.3037 - loss: 0.3880 - val_dice_coefficient: 0.4796 - val_iou: 0.3155 -
val_loss: 0.3914

Epoch 9/20

40/40 ————— 34s 863ms/step - dice_coefficient: 0.4726 - iou:
0.3097 - loss: 0.3866 - val_dice_coefficient: 0.4911 - val_iou: 0.3255 -
val_loss: 0.3816

Epoch 10/20

40/40 ————— 34s 860ms/step - dice_coefficient: 0.4680 - iou:
0.3063 - loss: 0.4156 - val_dice_coefficient: 0.4121 - val_iou: 0.2596 -
val_loss: 0.4510

Epoch 11/20

40/40 ————— 34s 843ms/step - dice_coefficient: 0.4301 - iou:
0.2743 - loss: 0.4345 - val_dice_coefficient: 0.4560 - val_iou: 0.2954 -
val_loss: 0.4137

Epoch 12/20

40/40 ————— 34s 848ms/step - dice_coefficient: 0.4411 - iou:
0.2836 - loss: 0.4148 - val_dice_coefficient: 0.4468 - val_iou: 0.2877 -
val_loss: 0.4140

Epoch 13/20

40/40 ————— 34s 846ms/step - dice_coefficient: 0.4418 - iou:
0.2837 - loss: 0.4180 - val_dice_coefficient: 0.4583 - val_iou: 0.2973 -
val_loss: 0.4125

Epoch 14/20

40/40 ————— 34s 842ms/step - dice_coefficient: 0.4312 - iou:
0.2756 - loss: 0.4349 - val_dice_coefficient: 0.4475 - val_iou: 0.2883 -
val_loss: 0.4184

Epoch 15/20

40/40 ————— 33s 817ms/step - dice_coefficient: 0.4506 - iou:
0.2910 - loss: 0.4129 - val_dice_coefficient: 0.4536 - val_iou: 0.2934 -
val_loss: 0.4095

Epoch 16/20

40/40 ————— 34s 842ms/step - dice_coefficient: 0.4523 - iou:
0.2924 - loss: 0.4101 - val_dice_coefficient: 0.4161 - val_iou: 0.2628 -
val_loss: 0.4260

Epoch 17/20

40/40 ————— 33s 819ms/step - dice_coefficient: 0.4229 - iou:
0.2699 - loss: 0.4302 - val_dice_coefficient: 0.4651 - val_iou: 0.3031 -
val_loss: 0.4088

Epoch 18/20

40/40 ————— 33s 818ms/step - dice_coefficient: 0.4529 - iou:
0.2928 - loss: 0.4106 - val_dice_coefficient: 4.0044e-04 - val_iou: 2.0026e-
04 - val_loss: 2.8609

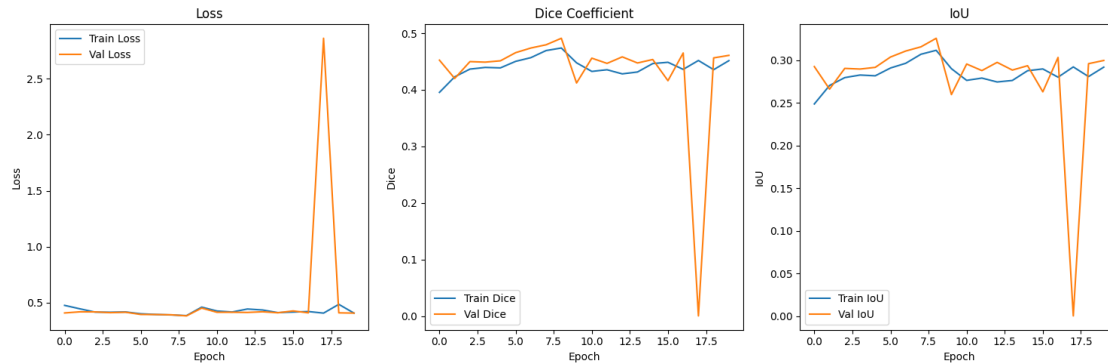
Epoch 19/20

40/40 ————— 33s 816ms/step - dice_coefficient: 0.3940 - iou:
0.2527 - loss: 0.6866 - val_dice_coefficient: 0.4565 - val_iou: 0.2958 -
val_loss: 0.4089

Epoch 20/20

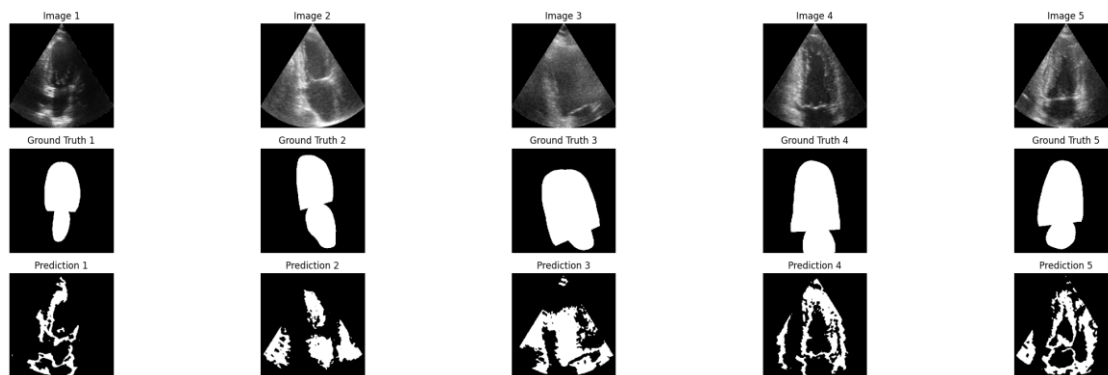
40/40 ————— 32s 813ms/step - dice_coefficient: 0.4506 - iou:

0.2908 - loss: 0.4065 - val_dice_coefficient: 0.4609 - val_iou: 0.2995 -
 val_loss: 0.4066
 Test Loss: 0.4027
 Test Dice Coefficient: 0.4462
 Test IoU: 0.2873



E0000 00:00:1754377790.617890 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.
 E0000 00:00:1754377790.843091 120 gpu_timer.cc:82] Delay kernel timed out: measured time has sub-optimal accuracy. There may be a missing warmup execution, please investigate in Nsight Systems.

1/1 ————— 13s 13s/step



```
def unet_plus_plus(input_shape=(256, 256, 1)):
    inputs = layers.Input(input_shape)

    c1 = layers.Conv2D(64, 3, activation='relu', padding='same')(inputs)
    c1 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1)
    p1 = layers.MaxPooling2D((2, 2))(c1)

    c2 = layers.Conv2D(128, 3, activation='relu', padding='same')(p1)
    c2 = layers.Conv2D(128, 3, activation='relu', padding='same')(c2)
    p2 = layers.MaxPooling2D((2, 2))(c2)

    c3 = layers.Conv2D(256, 3, activation='relu', padding='same')(p2)
```

```

c3 = layers.Conv2D(256, 3, activation='relu', padding='same')(c3)
p3 = layers.MaxPooling2D((2, 2))(c3)

c4 = layers.Conv2D(512, 3, activation='relu', padding='same')(p3)
c4 = layers.Conv2D(512, 3, activation='relu', padding='same')(c4)
p4 = layers.MaxPooling2D((2, 2))(c4)

c5 = layers.Conv2D(1024, 3, activation='relu', padding='same')(p4)
c5 = layers.Conv2D(1024, 3, activation='relu', padding='same')(c5)

u4_1 = layers.Conv2DTranspose(512, 2, strides=(2, 2), padding='same')(c5)
u4_1 = layers.concatenate([u4_1, c4])
c4_1 = layers.Conv2D(512, 3, activation='relu', padding='same')(u4_1)
c4_1 = layers.Conv2D(512, 3, activation='relu', padding='same')(c4_1)

u3_1 = layers.Conv2DTranspose(256, 2, strides=(2, 2), padding='same')(c4)
u3_1 = layers.concatenate([u3_1, c3])
c3_1 = layers.Conv2D(256, 3, activation='relu', padding='same')(u3_1)
c3_1 = layers.Conv2D(256, 3, activation='relu', padding='same')(c3_1)

u3_2 = layers.Conv2DTranspose(256, 2, strides=(2, 2),
padding='same')(c4_1)
u3_2 = layers.concatenate([u3_2, c3, c3_1])
c3_2 = layers.Conv2D(256, 3, activation='relu', padding='same')(u3_2)
c3_2 = layers.Conv2D(256, 3, activation='relu', padding='same')(c3_2)

u2_1 = layers.Conv2DTranspose(128, 2, strides=(2, 2), padding='same')(c3)
u2_1 = layers.concatenate([u2_1, c2])
c2_1 = layers.Conv2D(128, 3, activation='relu', padding='same')(u2_1)
c2_1 = layers.Conv2D(128, 3, activation='relu', padding='same')(c2_1)

u2_2 = layers.Conv2DTranspose(128, 2, strides=(2, 2),
padding='same')(c3_1)
u2_2 = layers.concatenate([u2_2, c2, c2_1])
c2_2 = layers.Conv2D(128, 3, activation='relu', padding='same')(u2_2)
c2_2 = layers.Conv2D(128, 3, activation='relu', padding='same')(c2_2)

u2_3 = layers.Conv2DTranspose(128, 2, strides=(2, 2),
padding='same')(c3_2)
u2_3 = layers.concatenate([u2_3, c2, c2_1, c2_2])
c2_3 = layers.Conv2D(128, 3, activation='relu', padding='same')(u2_3)
c2_3 = layers.Conv2D(128, 3, activation='relu', padding='same')(c2_3)

u1_1 = layers.Conv2DTranspose(64, 2, strides=(2, 2), padding='same')(c2)
u1_1 = layers.concatenate([u1_1, c1])
c1_1 = layers.Conv2D(64, 3, activation='relu', padding='same')(u1_1)
c1_1 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1_1)

u1_2 = layers.Conv2DTranspose(64, 2, strides=(2, 2),

```

```

padding='same')(c2_1)
    u1_2 = layers.concatenate([u1_2, c1, c1_1])
    c1_2 = layers.Conv2D(64, 3, activation='relu', padding='same')(u1_2)
    c1_2 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1_2)

    u1_3 = layers.Conv2DTranspose(64, 2, strides=(2, 2),
padding='same')(c2_2)
    u1_3 = layers.concatenate([u1_3, c1, c1_1, c1_2])
    c1_3 = layers.Conv2D(64, 3, activation='relu', padding='same')(u1_3)
    c1_3 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1_3)

    u1_4 = layers.Conv2DTranspose(64, 2, strides=(2, 2),
padding='same')(c2_3)
    u1_4 = layers.concatenate([u1_4, c1, c1_1, c1_2, c1_3])
    c1_4 = layers.Conv2D(64, 3, activation='relu', padding='same')(u1_4)
    c1_4 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1_4)

    outputs = layers.Conv2D(1, 1, activation='sigmoid')(c1_4)

    model = models.Model(inputs, outputs)
    return model

def load_and_preprocess_data(image_dir, mask_dir, img_size=(256, 256)):
    image_files = [f for f in os.listdir(image_dir) if f.endswith('.png')]
    mask_files = [f for f in os.listdir(mask_dir) if f.endswith('.png')]
    print(f"Found {len(image_files)} image files in {image_dir}")
    print(f"Found {len(mask_files)} mask files in {mask_dir}")
    image_basenames = {os.path.splitext(f)[0].replace('frame_', ''): f for f
in image_files}
    mask_basenames = {os.path.splitext(f)[0].replace('mask_', ''): f for f in
mask_files}
    common_keys = set(image_basenames.keys()) & set(mask_basenames.keys())
    print(f"Found {len(common_keys)} matching image-mask pairs")

    images = []
    masks = []

    for key in sorted(common_keys):
        img_path = os.path.join(image_dir, image_basenames[key])
        mask_path = os.path.join(mask_dir, mask_basenames[key])
        img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
        mask = cv2.imread(mask_path, cv2.IMREAD_GRAYSCALE)
        img = cv2.resize(img, img_size)
        mask = cv2.resize(mask, img_size)
        img = img.astype(np.float32) / 255.0
        mask = (mask > 0).astype(np.float32)
        images.append(img)
        masks.append(mask)
    images = np.array(images)[..., np.newaxis]

```

```

    masks = np.array(masks)[..., np.newaxis]
    return images, masks

def plot_training_history(history):
    fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(15, 5))
    ax1.plot(history.history['loss'], label='Train Loss')
    ax1.plot(history.history['val_loss'], label='Val Loss')
    ax1.set_title('Loss')
    ax1.set_xlabel('Epoch')
    ax1.set_ylabel('Loss')
    ax1.legend()

    ax2.plot(history.history['dice_coefficient'], label='Train Dice')
    ax2.plot(history.history['val_dice_coefficient'], label='Val Dice')
    ax2.set_title('Dice Coefficient')
    ax2.set_xlabel('Epoch')
    ax2.set_ylabel('Dice')
    ax2.legend()

    ax3.plot(history.history['iou'], label='Train IoU')
    ax3.plot(history.history['val_iou'], label='Val IoU')
    ax3.set_title('IoU')
    ax3.set_xlabel('Epoch')
    ax3.set_ylabel('IoU')
    ax3.legend()
    plt.tight_layout()
    plt.show()

def display_predictions(model, x_test, y_test, num_display=5):
    num_display = min(num_display, len(x_test))
    if num_display == 0:
        print("No test samples to display.")
        return

    preds = model.predict(x_test[:num_display])
    preds = (preds > 0.5).astype(np.float32)
    fig, axes = plt.subplots(3, num_display, figsize=(5 * num_display, 7))

    if num_display == 1:
        axes = [[axes[0]], [axes[1]], [axes[2]]]

    for i in range(num_display):
        axes[0][i].imshow(x_test[i, :, :, 0], cmap='gray')
        axes[0][i].set_title(f'Image {i+1}')
        axes[0][i].axis('off')
        axes[1][i].imshow(y_test[i, :, :, 0], cmap='gray')
        axes[1][i].set_title(f'Ground Truth {i+1}')
        axes[1][i].axis('off')
        axes[2][i].imshow(preds[i, :, :, 0], cmap='gray')

```

```

        axes[2][i].set_title(f'Prediction {i+1}')
        axes[2][i].axis('off')
    plt.tight_layout()
    plt.show()

image_dir = '/kaggle/input/camus-echocardiography-image-dataset/frames/'
mask_dir = '/kaggle/input/camus-echocardiography-image-dataset/masks/'

images, masks = load_and_preprocess_data(image_dir, mask_dir)

x_train, x_temp, y_train, y_temp = train_test_split(images, masks,
test_size=0.3, random_state=42)
x_val, x_test, y_val, y_test = train_test_split(x_temp, y_temp,
test_size=0.5, random_state=42)

print(f"Train: {len(x_train)}, Validation: {len(x_val)}, Test:
{len(x_test)}")
model = unet_plus_plus()

model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=[dice_coefficient, iou])

history = model.fit(x_train, y_train, validation_data=(x_val, y_val),
epochs=20, batch_size=16, verbose=1)

test_scores = model.evaluate(x_test, y_test, verbose=0)
print(f"Test Loss: {test_scores[0]:.4f}")
print(f"Test Dice Coefficient: {test_scores[1]:.4f}")
print(f"Test IoU: {test_scores[2]:.4f}")

plot_training_history(history)

display_predictions(model, x_test, y_test, num_display=5)

Found 900 image files in /kaggle/input/camus-echocardiography-image-
dataset/frames/
Found 900 mask files in /kaggle/input/camus-echocardiography-image-
dataset/masks/
Found 900 matching image-mask pairs
Train: 630, Validation: 135, Test: 135
Epoch 1/20

2025-08-05 07:36:54.035097: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=1,k3=0} for conv (f32[16,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
call(f32[16,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_

```


result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_schedule":false,"operation_queue_id":"0","wait_on_operation_queues":[] is taking a while...

2025-08-05 07:36:54.089966: E

external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation took 1.054971515s

Trying algorithm eng11{k2=1,k3=0} for conv (f32[16,64,256,256]{3,2,1,0}, u8[0]{0}) custom-call(f32[16,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0}, f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01, custom_call_target="__cudnn\$convBiasActivationForward", backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_schedule":false,"operation_queue_id":"0","wait_on_operation_queues":[] is taking a while...

2025-08-05 07:37:47.134192: E

external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm eng20{k2=1,k3=0} for conv (f32[64,320,3,3]{3,2,1,0}, u8[0]{0}) custom-call(f32[16,320,256,256]{3,2,1,0}, f32[16,64,256,256]{3,2,1,0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01, custom_call_target="__cudnn\$convBackwardFilter", backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_schedule":false,"operation_queue_id":"0","wait_on_operation_queues":[] is taking a while...

2025-08-05 07:37:47.179391: E

external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation took 1.045310262s

Trying algorithm eng20{k2=1,k3=0} for conv (f32[64,320,3,3]{3,2,1,0}, u8[0]{0}) custom-call(f32[16,320,256,256]{3,2,1,0}, f32[16,64,256,256]{3,2,1,0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01, custom_call_target="__cudnn\$convBackwardFilter", backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_schedule":false,"operation_queue_id":"0","wait_on_operation_queues":[] is taking a while...

2025-08-05 07:37:48.179625: E

external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm eng1{k2=2,k3=0} for conv (f32[64,320,3,3]{3,2,1,0}, u8[0]{0}) custom-call(f32[16,320,256,256]{3,2,1,0}, f32[16,64,256,256]{3,2,1,0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01, custom_call_target="__cudnn\$convBackwardFilter", backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_schedule":false,"operation_queue_id":"0","wait_on_operation_queues":[] is taking a while...

2025-08-05 07:37:48.279180: E

external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation took 1.099652719s

Trying algorithm eng1{k2=2,k3=0} for conv (f32[64,320,3,3]{3,2,1,0}, u8[0]{0}) custom-call(f32[16,320,256,256]{3,2,1,0},

f32[16,64,256,256]{3,2,1,0}), window={size=3x3 pad=1_1x1_1},
dim_labels=bf01_oi01->bf01, custom_call_target="__cudnn\$convBackwardFilter",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kNone","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]}] is
taking a while...

40/40 ————— 0s 3s/step - dice_coefficient: 0.3782 - iou:
0.2351 - loss: 0.5217

2025-08-05 07:40:17.226253: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=1,k3=0} for conv (f32[16,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
call(f32[16,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]}] is
taking a while...

2025-08-05 07:40:17.242628: E
external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.016563687s
Trying algorithm eng11{k2=1,k3=0} for conv (f32[16,64,256,256]{3,2,1,0},
u8[0]{0}) custom-call(f32[16,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0},
f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]}] is
taking a while...

40/40 ————— 258s 4s/step - dice_coefficient: 0.3791 - iou:
0.2357 - loss: 0.5200 - val_dice_coefficient: 0.4594 - val_iou: 0.2983 -
val_loss: 0.4064

Epoch 2/20

40/40 ————— 91s 2s/step - dice_coefficient: 0.4514 - iou:
0.2919 - loss: 0.4032 - val_dice_coefficient: 0.4586 - val_iou: 0.2976 -
val_loss: 0.4110

Epoch 3/20

40/40 ————— 90s 2s/step - dice_coefficient: 0.4383 - iou:
0.2835 - loss: 0.4348 - val_dice_coefficient: 0.4458 - val_iou: 0.2869 -
val_loss: 0.4060

Epoch 4/20

40/40 ————— 90s 2s/step - dice_coefficient: 0.4526 - iou:
0.2926 - loss: 0.4056 - val_dice_coefficient: 0.4504 - val_iou: 0.2907 -
val_loss: 0.3977

Epoch 5/20

40/40 ————— 90s 2s/step - dice_coefficient: 0.4777 - iou:
0.3148 - loss: 0.3893 - val_dice_coefficient: 0.5294 - val_iou: 0.3601 -

val_loss: 0.3715
Epoch 6/20
40/40 —————90s 2s/step - dice_coefficient: 0.5508 - iou:
0.3831 - loss: 0.3500 - val_dice_coefficient: 0.6478 - val_iou: 0.4791 -
val_loss: 0.3052
Epoch 7/20
40/40 —————90s 2s/step - dice_coefficient: 0.6923 - iou:
0.5308 - loss: 0.2540 - val_dice_coefficient: 0.7086 - val_iou: 0.5488 -
val_loss: 0.2190
Epoch 8/20
40/40 —————90s 2s/step - dice_coefficient: 0.7429 - iou:
0.5916 - loss: 0.2087 - val_dice_coefficient: 0.7812 - val_iou: 0.6411 -
val_loss: 0.1816
Epoch 9/20
40/40 —————90s 2s/step - dice_coefficient: 0.7882 - iou:
0.6510 - loss: 0.1787 - val_dice_coefficient: 0.7679 - val_iou: 0.6235 -
val_loss: 0.2211
Epoch 10/20
40/40 —————90s 2s/step - dice_coefficient: 0.8112 - iou:
0.6835 - loss: 0.1576 - val_dice_coefficient: 0.8375 - val_iou: 0.7206 -
val_loss: 0.1270
Epoch 11/20
40/40 —————90s 2s/step - dice_coefficient: 0.8530 - iou:
0.7442 - loss: 0.1211 - val_dice_coefficient: 0.8741 - val_iou: 0.7765 -
val_loss: 0.1141
Epoch 12/20
40/40 —————90s 2s/step - dice_coefficient: 0.8716 - iou:
0.7726 - loss: 0.1073 - val_dice_coefficient: 0.8531 - val_iou: 0.7439 -
val_loss: 0.1283
Epoch 13/20
40/40 —————90s 2s/step - dice_coefficient: 0.8692 - iou:
0.7690 - loss: 0.1049 - val_dice_coefficient: 0.8960 - val_iou: 0.8116 -
val_loss: 0.0987
Epoch 14/20
40/40 —————90s 2s/step - dice_coefficient: 0.8962 - iou:
0.8121 - loss: 0.0868 - val_dice_coefficient: 0.8805 - val_iou: 0.7867 -
val_loss: 0.1034
Epoch 15/20
40/40 —————90s 2s/step - dice_coefficient: 0.8990 - iou:
0.8167 - loss: 0.0829 - val_dice_coefficient: 0.8929 - val_iou: 0.8067 -
val_loss: 0.0950
Epoch 16/20
40/40 —————91s 2s/step - dice_coefficient: 0.9040 - iou:
0.8250 - loss: 0.0813 - val_dice_coefficient: 0.8996 - val_iou: 0.8177 -
val_loss: 0.0896
Epoch 17/20
40/40 —————90s 2s/step - dice_coefficient: 0.9072 - iou:
0.8302 - loss: 0.0766 - val_dice_coefficient: 0.9108 - val_iou: 0.8363 -
val_loss: 0.0895
Epoch 18/20

40/40 —————91s 2s/step - dice_coefficient: 0.9150 - iou:
0.8434 - loss: 0.0711 - val_dice_coefficient: 0.9135 - val_iou: 0.8408 -
val_loss: 0.0855

Epoch 19/20

40/40 —————90s 2s/step - dice_coefficient: 0.9210 - iou:
0.8536 - loss: 0.0639 - val_dice_coefficient: 0.9080 - val_iou: 0.8316 -
val_loss: 0.0821

Epoch 20/20

40/40 —————90s 2s/step - dice_coefficient: 0.9228 - iou:
0.8567 - loss: 0.0621 - val_dice_coefficient: 0.9130 - val_iou: 0.8400 -
val_loss: 0.0830

2025-08-05 08:09:27.378055: E

external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=3,k3=0} for conv (f32[32,128,128,128]{3,2,1,0}, u8[0]{0}) custom-
call(f32[32,512,128,128]{3,2,1,0}, f32[128,512,3,3]{3,2,1,0}, f32[128]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...

2025-08-05 08:09:27.383507: E

external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.00557485s

Trying algorithm eng11{k2=3,k3=0} for conv (f32[32,128,128,128]{3,2,1,0},
u8[0]{0}) custom-call(f32[32,512,128,128]{3,2,1,0},
f32[128,512,3,3]{3,2,1,0}, f32[128]{0}), window={size=3x3 pad=1_1x1_1},
dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...

2025-08-05 08:09:32.904974: E

external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=1,k3=0} for conv (f32[32,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
call(f32[32,192,256,256]{3,2,1,0}, f32[64,192,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn\$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...

2025-08-05 08:09:33.124655: E

external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.219846952s

Trying algorithm eng11{k2=1,k3=0} for conv (f32[32,64,256,256]{3,2,1,0},
u8[0]{0}) custom-call(f32[32,192,256,256]{3,2,1,0}, f32[64,192,3,3]{3,2,1,0},
f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,

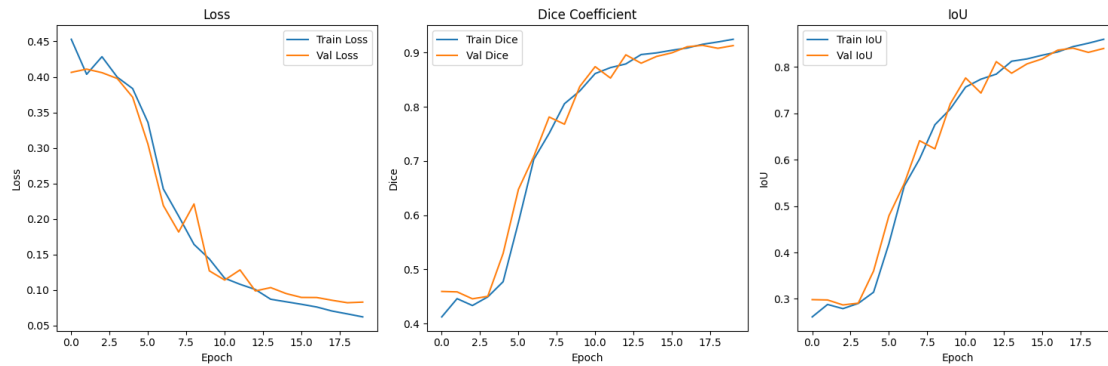
```
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]} is
taking a while...
2025-08-05 08:09:39.048911: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=3,k3=0} for conv (f32[32,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
call(f32[32,256,256,256]{3,2,1,0}, f32[64,256,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]} is
taking a while...
2025-08-05 08:09:39.063430: E
external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.014636754s
Trying algorithm eng11{k2=3,k3=0} for conv (f32[32,64,256,256]{3,2,1,0},
u8[0]{0}) custom-call(f32[32,256,256,256]{3,2,1,0}, f32[64,256,3,3]{3,2,1,0},
f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]} is
taking a while...
2025-08-05 08:09:40.063722: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=1,k3=0} for conv (f32[32,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
call(f32[32,256,256,256]{3,2,1,0}, f32[64,256,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]} is
taking a while...
2025-08-05 08:09:40.681034: E
external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.617519247s
Trying algorithm eng11{k2=1,k3=0} for conv (f32[32,64,256,256]{3,2,1,0},
u8[0]{0}) custom-call(f32[32,256,256,256]{3,2,1,0}, f32[64,256,3,3]{3,2,1,0},
f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[]} is
taking a while...
2025-08-05 08:09:47.804795: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=3,k3=0} for conv (f32[32,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
```

```

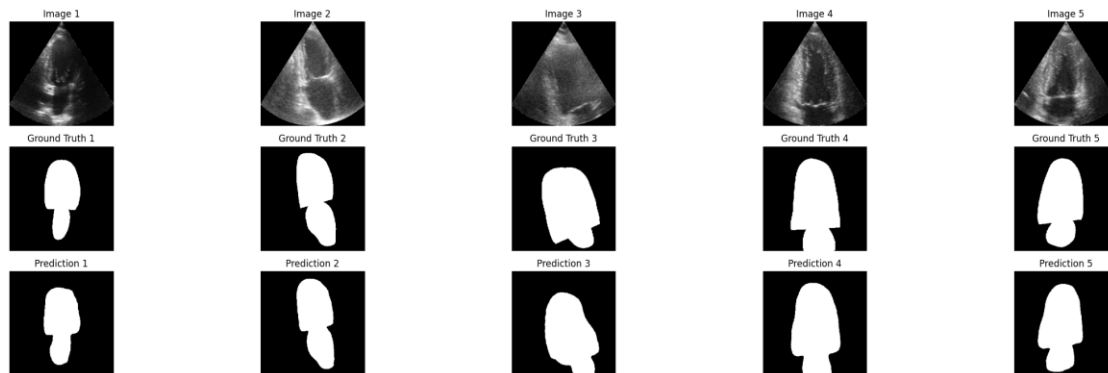
call(f32[32,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...
2025-08-05 08:09:48.066440: E
external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 1.261816178s
Trying algorithm eng11{k2=3,k3=0} for conv (f32[32,64,256,256]{3,2,1,0},
u8[0]{0}) custom-call(f32[32,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0},
f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...
2025-08-05 08:09:49.066705: E
external/local_xla/xla/service/slow_operation_alarm.cc:65] Trying algorithm
eng11{k2=1,k3=0} for conv (f32[32,64,256,256]{3,2,1,0}, u8[0]{0}) custom-
call(f32[32,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0}, f32[64]{0}),
window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...
2025-08-05 08:09:50.078371: E
external/local_xla/xla/service/slow_operation_alarm.cc:133] The operation
took 2.0118464s
Trying algorithm eng11{k2=1,k3=0} for conv (f32[32,64,256,256]{3,2,1,0},
u8[0]{0}) custom-call(f32[32,320,256,256]{3,2,1,0}, f32[64,320,3,3]{3,2,1,0},
f32[64]{0}), window={size=3x3 pad=1_1x1_1}, dim_labels=bf01_oi01->bf01,
custom_call_target="__cudnn$convBiasActivationForward",
backend_config={"cudnn_conv_backend_config":{"activation_mode":"kRelu","conv_
result_scale":1,"leakyrelu_alpha":0,"side_input_scale":0},"force_earliest_sch
edule":false,"operation_queue_id":"0","wait_on_operation_queues":[] } is
taking a while...

Test Loss: 0.0721
Test Dice Coefficient: 0.9156
Test IoU: 0.8444

```



1/1 ————— 10s 10s/step



```
def inception_block(inputs, filters):
    conv1x1 = layers.Conv2D(filters // 4, 1, activation='relu',
padding='same')(inputs)

    conv3x3 = layers.Conv2D(filters // 4, 1, activation='relu',
padding='same')(inputs)
    conv3x3 = layers.Conv2D(filters // 4, 3, activation='relu',
padding='same')(conv3x3)

    conv5x5 = layers.Conv2D(filters // 4, 1, activation='relu',
padding='same')(inputs)
    conv5x5 = layers.Conv2D(filters // 4, 5, activation='relu',
padding='same')(conv5x5)

    pool = layers.MaxPooling2D(3, strides=1, padding='same')(inputs)
    pool = layers.Conv2D(filters // 4, 1, activation='relu',
padding='same')(pool)

    output = layers.concatenate([conv1x1, conv3x3, conv5x5, pool])
    return output

def inception_unet_plus_plus(input_shape=(128, 128, 1)):
    inputs = layers.Input(input_shape)
```

```

c1 = inception_block(inputs, 32)
c1 = inception_block(c1, 32)
p1 = layers.MaxPooling2D((2, 2))(c1)

c2 = inception_block(p1, 64)
c2 = inception_block(c2, 64)
p2 = layers.MaxPooling2D((2, 2))(c2)

c3 = inception_block(p2, 128)
c3 = inception_block(c3, 128)
p3 = layers.MaxPooling2D((2, 2))(c3)

c4 = inception_block(p3, 256)
c4 = inception_block(c4, 256)
p4 = layers.MaxPooling2D((2, 2))(c4)

c5 = inception_block(p4, 512)
c5 = inception_block(c5, 512)

u4_1 = layers.Conv2DTranspose(256, 2, strides=(2, 2), padding='same')(c5)
u4_1 = layers.concatenate([u4_1, c4])
c4_1 = inception_block(u4_1, 256)
c4_1 = inception_block(c4_1, 256)

u3_1 = layers.Conv2DTranspose(128, 2, strides=(2, 2), padding='same')(c4)
u3_1 = layers.concatenate([u3_1, c3])
c3_1 = inception_block(u3_1, 128)
c3_1 = inception_block(c3_1, 128)

u3_2 = layers.Conv2DTranspose(128, 2, strides=(2, 2),
padding='same')(c4_1)
u3_2 = layers.concatenate([u3_2, c3, c3_1])
c3_2 = inception_block(u3_2, 128)
c3_2 = inception_block(c3_2, 128)

u2_1 = layers.Conv2DTranspose(64, 2, strides=(2, 2), padding='same')(c3)
u2_1 = layers.concatenate([u2_1, c2])
c2_1 = inception_block(u2_1, 64)
c2_1 = inception_block(c2_1, 64)

u2_2 = layers.Conv2DTranspose(64, 2, strides=(2, 2),
padding='same')(c3_1)
u2_2 = layers.concatenate([u2_2, c2, c2_1])
c2_2 = inception_block(u2_2, 64)
c2_2 = inception_block(c2_2, 64)

u2_3 = layers.Conv2DTranspose(64, 2, strides=(2, 2),
padding='same')(c3_2)
u2_3 = layers.concatenate([u2_3, c2, c2_1, c2_2])

```



```

c2_3 = inception_block(u2_3, 64)
c2_3 = inception_block(c2_3, 64)

u1_1 = layers.Conv2DTranspose(32, 2, strides=(2, 2), padding='same')(c2)
u1_1 = layers.concatenate([u1_1, c1])
c1_1 = inception_block(u1_1, 32)
c1_1 = inception_block(c1_1, 32)

u1_2 = layers.Conv2DTranspose(32, 2, strides=(2, 2),
padding='same')(c2_1)
u1_2 = layers.concatenate([u1_2, c1, c1_1])
c1_2 = inception_block(u1_2, 32)
c1_2 = inception_block(c1_2, 32)

u1_3 = layers.Conv2DTranspose(32, 2, strides=(2, 2),
padding='same')(c2_2)
u1_3 = layers.concatenate([u1_3, c1, c1_1, c1_2])
c1_3 = inception_block(u1_3, 32)
c1_3 = inception_block(c1_3, 32)

u1_4 = layers.Conv2DTranspose(32, 2, strides=(2, 2),
padding='same')(c2_3)
u1_4 = layers.concatenate([u1_4, c1, c1_1, c1_2, c1_3])
c1_4 = inception_block(u1_4, 32)
c1_4 = inception_block(c1_4, 32)

outputs = layers.Conv2D(1, 1, activation='sigmoid')(c1_4)

model = models.Model(inputs, outputs)
return model

def load_and_preprocess_data(image_dir, mask_dir, img_size=(128, 128)):
    image_files = [f for f in os.listdir(image_dir) if f.endswith('.png')]
    mask_files = [f for f in os.listdir(mask_dir) if f.endswith('.png')]
    print(f"Found {len(image_files)} image files in {image_dir}")
    print(f"Found {len(mask_files)} mask files in {mask_dir}")
    image_basenames = {os.path.splitext(f)[0].replace('frame_', ''): f for f
in image_files}
    mask_basenames = {os.path.splitext(f)[0].replace('mask_', ''): f for f
in mask_files}
    common_keys = set(image_basenames.keys()) & set(mask_basenames.keys())
    print(f"Found {len(common_keys)} matching image-mask pairs")

    images = []
    masks = []

    for key in sorted(common_keys):
        img_path = os.path.join(image_dir, image_basenames[key])
        mask_path = os.path.join(mask_dir, mask_basenames[key])

```

```

    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    mask = cv2.imread(mask_path, cv2.IMREAD_GRAYSCALE)
    img = cv2.resize(img, img_size)
    mask = cv2.resize(mask, img_size)
    img = img.astype(np.float32) / 255.0
    mask = (mask > 0).astype(np.float32)
    images.append(img)
    masks.append(mask)

images = np.array(images)[..., np.newaxis]
masks = np.array(masks)[..., np.newaxis]
return images, masks

def plot_training_history(history):
    fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(15, 5))
    ax1.plot(history.history['loss'], label='Train Loss')
    ax1.plot(history.history['val_loss'], label='Val Loss')
    ax1.set_title('Loss')
    ax1.set_xlabel('Epoch')
    ax1.set_ylabel('Loss')
    ax1.legend()
    ax2.plot(history.history['dice_coefficient'], label='Train Dice')
    ax2.plot(history.history['val_dice_coefficient'], label='Val Dice')
    ax2.set_title('Dice Coefficient')
    ax2.set_xlabel('Epoch')
    ax2.set_ylabel('Dice')
    ax2.legend()
    ax3.plot(history.history['iou'], label='Train IoU')
    ax3.plot(history.history['val_iou'], label='Val IoU')
    ax3.set_title('IoU')
    ax3.set_xlabel('Epoch')
    ax3.set_ylabel('IoU')
    ax3.legend()
    plt.tight_layout()
    plt.show()

def display_predictions(model, x_test, y_test, num_display=5):
    num_display = min(num_display, len(x_test))
    if num_display == 0:
        print("No test samples to display.")
        return
    preds = model.predict(x_test[:num_display])
    preds = (preds > 0.5).astype(np.float32)
    fig, axes = plt.subplots(3, num_display, figsize=(5 * num_display, 7))

    if num_display == 1:
        axes = [[axes[0]], [axes[1]], [axes[2]]]

    for i in range(num_display):

```

```

axes[0][i].imshow(x_test[i, :, :, 0], cmap='gray')
axes[0][i].set_title(f'Image {i+1}')
axes[0][i].axis('off')
axes[1][i].imshow(y_test[i, :, :, 0], cmap='gray')
axes[1][i].set_title(f'Ground Truth {i+1}')
axes[1][i].axis('off')
axes[2][i].imshow(preds[i, :, :, 0], cmap='gray')
axes[2][i].set_title(f'Prediction {i+1}')
axes[2][i].axis('off')
plt.tight_layout()
plt.show()

```

```

image_dir = '/kaggle/input/camus-echocardiography-image-dataset/frames/'
mask_dir = '/kaggle/input/camus-echocardiography-image-dataset/masks/'
images, masks = load_and_preprocess_data(image_dir, mask_dir)
x_train, x_temp, y_train, y_temp = train_test_split(images, masks,
test_size=0.3, random_state=42)
x_val, x_test, y_val, y_test = train_test_split(x_temp, y_temp,
test_size=0.5, random_state=42)
print(f"Train: {len(x_train)}, Validation: {len(x_val)}, Test:
{len(x_test)}")
model = inception_unet_plus_plus()
model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=[dice_coefficient, iou])
history = model.fit(x_train, y_train, validation_data=(x_val, y_val),
epochs=20, batch_size=8, verbose=1)
test_scores = model.evaluate(x_test, y_test, verbose=0)
print(f"Test Loss: {test_scores[0]:.4f}")
print(f"Test Dice Coefficient: {test_scores[1]:.4f}")
print(f"Test IoU: {test_scores[2]:.4f}")
plot_training_history(history)
display_predictions(model, x_test, y_test, num_display=5)

```

Found 900 image files in /kaggle/input/camus-echocardiography-image-dataset/frames/

Found 900 mask files in /kaggle/input/camus-echocardiography-image-dataset/masks/

Found 900 matching image-mask pairs

Train: 630, Validation: 135, Test: 135

Epoch 1/20

79/79 —————157s 744ms/step - dice_coefficient: 0.3895 - iou: 0.2492 - loss: 0.4886 - val_dice_coefficient: 0.6840 - val_iou: 0.5199 - val_loss: 0.2317

Epoch 2/20

79/79 —————14s 182ms/step - dice_coefficient: 0.7207 - iou: 0.5655 - loss: 0.2193 - val_dice_coefficient: 0.7852 - val_iou: 0.6465 - val_loss: 0.1929

Epoch 3/20

79/79 —————15s 185ms/step - dice_coefficient: 0.7834 - iou: 0.6449 - loss: 0.1749 - val_dice_coefficient: 0.8332 - val_iou: 0.7144 -

val_loss: 0.1355
Epoch 4/20
79/79 —————14s 180ms/step - dice_coefficient: 0.8419 - iou:
0.7276 - loss: 0.1270 - val_dice_coefficient: 0.8656 - val_iou: 0.7632 -
val_loss: 0.1157
Epoch 5/20
79/79 —————14s 177ms/step - dice_coefficient: 0.8727 - iou:
0.7744 - loss: 0.1051 - val_dice_coefficient: 0.8771 - val_iou: 0.7812 -
val_loss: 0.1013
Epoch 6/20
79/79 —————14s 177ms/step - dice_coefficient: 0.8779 - iou:
0.7827 - loss: 0.0994 - val_dice_coefficient: 0.9053 - val_iou: 0.8271 -
val_loss: 0.0966
Epoch 7/20
79/79 —————14s 178ms/step - dice_coefficient: 0.8855 - iou:
0.7951 - loss: 0.0956 - val_dice_coefficient: 0.8918 - val_iou: 0.8050 -
val_loss: 0.1037
Epoch 8/20
79/79 —————14s 179ms/step - dice_coefficient: 0.8919 - iou:
0.8052 - loss: 0.0902 - val_dice_coefficient: 0.9037 - val_iou: 0.8245 -
val_loss: 0.0818
Epoch 9/20
79/79 —————14s 181ms/step - dice_coefficient: 0.9093 - iou:
0.8337 - loss: 0.0744 - val_dice_coefficient: 0.9093 - val_iou: 0.8338 -
val_loss: 0.0839
Epoch 10/20
79/79 —————14s 180ms/step - dice_coefficient: 0.9143 - iou:
0.8425 - loss: 0.0704 - val_dice_coefficient: 0.9034 - val_iou: 0.8239 -
val_loss: 0.0833
Epoch 11/20
79/79 —————14s 179ms/step - dice_coefficient: 0.9128 - iou:
0.8397 - loss: 0.0720 - val_dice_coefficient: 0.9105 - val_iou: 0.8360 -
val_loss: 0.0809
Epoch 12/20
79/79 —————14s 178ms/step - dice_coefficient: 0.9231 - iou:
0.8573 - loss: 0.0621 - val_dice_coefficient: 0.9149 - val_iou: 0.8433 -
val_loss: 0.0818
Epoch 13/20
79/79 —————14s 178ms/step - dice_coefficient: 0.9255 - iou:
0.8614 - loss: 0.0610 - val_dice_coefficient: 0.9145 - val_iou: 0.8427 -
val_loss: 0.0724
Epoch 14/20
79/79 —————14s 179ms/step - dice_coefficient: 0.9242 - iou:
0.8591 - loss: 0.0612 - val_dice_coefficient: 0.9154 - val_iou: 0.8441 -
val_loss: 0.0729
Epoch 15/20
79/79 —————14s 180ms/step - dice_coefficient: 0.9253 - iou:
0.8610 - loss: 0.0577 - val_dice_coefficient: 0.9204 - val_iou: 0.8527 -
val_loss: 0.0728
Epoch 16/20

79/79 ————— 14s 179ms/step - dice_coefficient: 0.9294 - iou: 0.8681 - loss: 0.0545 - val_dice_coefficient: 0.9241 - val_iou: 0.8590 - val_loss: 0.0739

Epoch 17/20

79/79 ————— 14s 179ms/step - dice_coefficient: 0.9370 - iou: 0.8814 - loss: 0.0519 - val_dice_coefficient: 0.9227 - val_iou: 0.8566 - val_loss: 0.0737

Epoch 18/20

79/79 ————— 14s 180ms/step - dice_coefficient: 0.9302 - iou: 0.8697 - loss: 0.0566 - val_dice_coefficient: 0.9193 - val_iou: 0.8508 - val_loss: 0.0796

Epoch 19/20

79/79 ————— 14s 180ms/step - dice_coefficient: 0.9263 - iou: 0.8628 - loss: 0.0598 - val_dice_coefficient: 0.9213 - val_iou: 0.8543 - val_loss: 0.0716

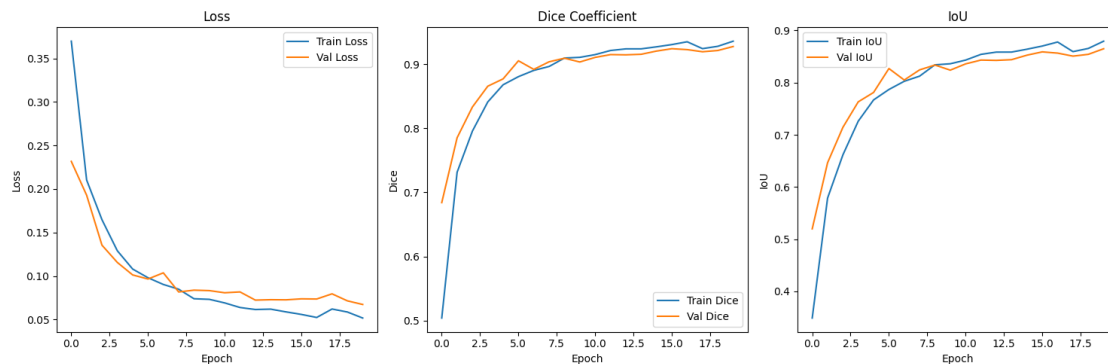
Epoch 20/20

79/79 ————— 14s 179ms/step - dice_coefficient: 0.9353 - iou: 0.8785 - loss: 0.0515 - val_dice_coefficient: 0.9275 - val_iou: 0.8650 - val_loss: 0.0674

Test Loss: 0.0609

Test Dice Coefficient: 0.9270

Test IoU: 0.8640



1/1 ————— 8s 8s/step

